Graphs 2 — Equation of a line · Sketching · Simultaneous equations graphically

Squared paper needed for questions that say *draw*; a *sketch* is a rough drawing showing the intercepts and slope. Marks in brackets. ★ = harder question.

A · FINDING THE EQUATION OF A LINE

1 Find the equation of the straight line through $A(-2,\ -7)$ and $B(2,\ 5)$. [3]

> Gradient $m = \dfrac{5-(-7)}{2-(-2)} = \dfrac{12}{4} = 3$. So $y = 3x + c$. Substitute B:
> $5 = 6 + c \Rightarrow c = -1$. Equation: $y = 3x - 1$. Check with A: $3(-2) - 1 = -7$ ✓

2 A straight line passes through the origin O and the point $A(4,\ -2)$. [4]

a Find the equation of OA.

b Find the equation of the line parallel to OA that passes through $(-3,\ 1)$.

> **a** Through the origin, so $c = 0$: $m = \dfrac{-2-0}{4-0} = -\dfrac{1}{2}$, giving $y = -\frac{1}{2}x$.
> **b** Parallel → same gradient: $y = -\frac{1}{2}x + c$. Substitute $(-3, 1)$: $1 = \frac{3}{2} + c \Rightarrow c = -\frac{1}{2}$.
> Equation: $y = -\frac{1}{2}x - \frac{1}{2}$.

B · SKETCHING

3 Sketch these graphs, marking where each line crosses the axes. [6]

a $y = 2x - 3$ **b** $3x + 4y = 24$ **c** $y = -x + 1$

> **a** Gradient 2 (steep, rising), y-intercept $(0, -3)$; crosses the x-axis where $2x - 3 = 0$, at $(1.5, 0)$.
> **b** $x = 0 \Rightarrow y = 6$; $y = 0 \Rightarrow x = 8$. Join $(0, 6)$ to $(8, 0)$: a falling line (gradient $-\frac{3}{4}$).
> **c** Gradient -1 (falling at 45°), y-intercept $(0, 1)$, x-intercept $(1, 0)$.

C · SIMULTANEOUS EQUATIONS GRAPHICALLY

4 Solve the simultaneous equations $x + y = 8$ and $y = 2x - 1$ graphically. Draw both axes from -2 to 8. [6]

> $x + y = 8$ passes through $(0, 8)$ and $(8, 0)$. $y = 2x - 1$ passes through $(0, -1)$, $(2, 3)$ and $(4, 7)$. The lines cross at $(3, 5)$, so $x = 3, y = 5$. Check: $3 + 5 = 8$ ✓ and $2 \times 3 - 1 = 5$ ✓

5 At a stationery shop, 3 pens and 2 pencils cost €8, while 1 pen and 4 pencils cost €6. Let a pen cost x euros and a pencil y euros. Write a pair of simultaneous equations and solve them graphically to find the cost of one pen and of one pencil. [6]

> Equations: $3x + 2y = 8$ and $x + 4y = 6$.
> First line: $(0, 4)$ and $\left(\frac{8}{3}, 0\right)$; second line: $(0, 1.5)$ and $(6, 0)$. They cross at $(2, 1)$: a pen costs **€2**, a pencil **€1**. Check: $6 + 2 = 8$ ✓, $2 + 4 = 6$ ✓

Total: 25 marks

Check: c comes from substituting a point the line passes through · parallel means equal gradients · the solution of simultaneous equations is the crossing point — substitute it back into *both* equations.